

Money & Banking

CHAPTER 01

Why Study Money, Banking and Financial Markets?

1. Why Study Financial Markets?

Financial markets are the backbone of a modern economy. They channel funds from people who have saved surplus money to people who have productive uses for it. This process of transferring funds from savers to borrowers is called financial intermediation, and it is crucial for economic efficiency and growth.

Financial Markets

Markets in which funds are transferred from people who have an excess of available funds (savers/lenders) to people who have a shortage (borrowers/investors). They improve economic efficiency by channeling funds to their most productive uses.

Example:

A young engineer has saved Rs. 500,000 but no business idea. A startup founder has a brilliant app concept but no capital. The bond or stock market acts as a bridge — the engineer buys bonds or shares issued by the startup, and the startup uses that money to hire developers and grow.

Why financial markets matter to YOU?

1. **Personal wealth:** Stock market booms make people richer and more confident; crashes wipe out savings. The 2008 financial crisis destroyed \$10+ trillion in U.S. household wealth.
2. **Economic growth:** Countries with well-developed financial markets consistently grow faster. Poorly functioning markets trap economies in poverty cycles.
3. **Business decisions:** Whether a company can expand, hire, or invest depends on its ability to raise funds cheaply in financial markets.
4. **Consumer spending:** Low interest rates make mortgages, car loans, and credit cards cheaper → consumers borrow and spend more → GDP grows.

Key Insight:

The three most important financial markets are:

1. The **Bond Market** (determines interest rates).
2. The **Stock Market** (affects business investment and consumer wealth).
3. The **Foreign Exchange Market** (affects trade and global competitiveness). All three are covered in this chapter.

2. The Bond Market and Interest Rates

1. Security (Financial Instrument)

A claim on the issuer's future income or assets. Any financial claim or piece of property that is subject to ownership is called a security (also known as a financial instrument). It can be bought and sold in financial markets.

Example: A Pakistan Investment Bond (PIB) is a security. When you buy it, the government owes you money. You can hold it until maturity or sell it to another investor in the secondary market.

2. Bond

A debt security — a promise by the issuer (borrower) to make periodic payments (interest) to the buyer (lender) for a specified period of time, and then repay the principal at maturity. Governments and corporations issue bonds to raise long-term funds.

Example: PTCL needs Rs. 5 billion to upgrade its network. It issues 10-year bonds at 12% annual interest. Investors lend Rs. 5 billion today; PTCL pays Rs. 600 million every year and returns the Rs. 5 billion after 10 years. This is a bond.

3. Interest Rate

The cost of borrowing money, expressed as a percentage of the amount borrowed per year. It is the price paid for renting funds. When interest rates rise, borrowing becomes expensive; when they fall, borrowing is cheap and economic activity tends to increase.

Example: If a home loan has a 10% interest rate, borrowing Rs. 10,000,000 costs Rs. 1,000,000 per year in interest. At 6%, it costs only Rs. 600,000 — making home ownership far more affordable. This is why interest rate cuts stimulate housing markets.

Real-world interest rate patterns (U.S. historical data):

Year / Period	3-Month T-Bill	U.S. Government Bond	Baa Corporate Bond
Early 1950s	~1.5%	~2.5%	~3.5%
1981 (peak)	~16%	~14%	~17%
1992–93	~3%	~6%	~8.5%
2003 (low)	~1%	~4%	~6%
2023 (high)	~5.3%	~4.6%	~5.7%

Note: Baa corporate bonds always carry the highest rate because they carry more default risk than government bonds. Treasury bills fluctuate the most because they are most sensitive to short-term monetary policy.

Key takeaways:

1. High interest rates discourage home buying, car purchases, and business investment — all of which require borrowing.
2. High interest rates encourage saving because bank deposits earn more.
3. Different bonds carry different rates. More risk = higher rate demanded by investors.
4. Interest rates are arguably the most watched variables in economics — the Federal Reserve's interest rate decisions dominate financial news worldwide.

3. The Stock Market

1. Common Stock (Equity)

A share of ownership in a corporation. Common stockholders are residual claimants — they receive dividends from profits (if any are paid) and any remaining assets if the company is liquidated. Unlike bonds, stocks do not promise fixed payments — returns depend on company performance.

Example: When you buy one share of Habib Bank Limited (HBL) on the Pakistan Stock Exchange (PSX), you own a tiny fraction of HBL. If HBL earns high profits, you may receive dividends. If HBL's prospects improve, its share price rises and you gain capital appreciation.

2. Stock Market

The market where shares of corporations are bought and sold. It is the most widely followed financial market because it directly reflects expectations about corporate profitability and the overall health of the economy. It is often called simply 'the market.'

Example: The Karachi Stock Exchange (KSE-100) is Pakistan's main stock market. When investor confidence is high (e.g., stable inflation, strong GDP growth), stock prices rise. When there is political uncertainty or economic crisis, the index falls sharply.

Why stock market performance matters:

3. **Wealth effect:** When stock prices rise, households feel wealthier □ spend more □ boosts GDP. This is why a 30% stock market crash triggers recessions even for people who don't own stocks (their employers might!).
4. **Business investment:** Higher share prices mean companies can raise more money by issuing new shares □ fund expansion, R&D, new factories.
5. **Retirement savings:** Most pension funds invest heavily in stocks. A stock market crash directly reduces future retirement income for millions of workers.

4. The Foreign Exchange Market

1. Foreign Exchange Market

The market where one country's currency is exchanged for another's. It operates 24 hours a day globally and is the largest financial market in the world by trading volume (over \$7

trillion per day). It determines the foreign exchange rate.

Example: To import a Japanese car worth ¥3,000,000, a Pakistani importer must first convert Pakistani rupees into Japanese yen using the foreign exchange market. If the exchange rate is PKR 0.60 per yen, the car costs PKR 1,800,000.

2. Exchange Rate

The price of one country's currency in terms of another country's currency. When a currency depreciates (weakens), imports become more expensive and exports become cheaper for foreigners. When a currency appreciates (strengthens), imports become cheaper but exports become more expensive.

Example: If 1 USD = PKR 250 and you want to import goods worth \$10,000, you need PKR 2,500,000. If the rupee depreciates to 1 USD = PKR 300, the same goods now cost PKR 3,000,000 — 20% more expensive due purely to currency movement.

3. The International Financial System (Simple Explanation)

The International Financial System refers to the movement of money and capital between countries, which affects domestic economies. When capital flows increase between countries, it influences exchange rates, investment, trade, and monetary policy. For example, if foreign investors invest in Pakistan, businesses grow and the economy improves, but if investors withdraw money, it can cause economic problems. The international financial system also involves decisions about fixed exchange rates, capital controls, and the role of international institutions like the IMF, which help countries maintain financial stability.

Strong vs. Weak currency — who wins and who loses?

Factor	Strong Currency Effect	Weak Currency Effect
Imports	Cheaper for domestic buyers	More expensive
Exports	More expensive for foreigners	Cheaper — more competitive
Inflation	Reduced (cheaper imports)	Increased (costlier imports)
Jobs in export industries	Hurt — loses competitiveness	Helped — gains orders
Tourism abroad	Cheaper — good for tourists	Expensive — reduces travel
Foreign tourists	Fewer visitors (expensive)	More visitors (affordable)

5. Why Study Banking and Financial Institutions?

Financial intermediaries are the plumbing of the economy. Without them, it would be nearly impossible for savers and borrowers to find each other efficiently. Understanding how banks and other institutions work — and how they can fail — is essential for understanding modern economies.

1. Financial Intermediaries

Institutions that borrow funds from people with surplus money (savers) and lend to those who need funds (borrowers/investors). They reduce transaction costs, manage risk, and solve information problems between savers and borrowers. They include banks, insurance companies, mutual funds, pension funds, and finance companies.

Example: Without banks, a person with Rs. 50,000 savings would need to personally find someone who wants to borrow exactly Rs. 50,000 for exactly the right period. Banks pool millions of small deposits and lend them out as mortgages, business loans, and car loans — creating enormous economic efficiency.

2. Banks (Depository Institutions)

Financial institutions that accept deposits from the public and use those funds to make loans. Banks are the most important type of financial intermediary. Types include: commercial banks (serve businesses and individuals), savings and loan associations (focus on mortgages), mutual savings banks, and credit unions (member-owned).

Example: HBL (Habib Bank Limited) accepts your salary deposit at 5% interest. It then lends that money to a business at 12% interest. The 7% difference (the 'spread') is the bank's profit for performing the intermediation service and taking on credit risk.

3. Financial Innovation and E-Finance

The development of new financial products and delivery methods, driven by advances in information technology. E-finance refers specifically to the delivery of financial services electronically. This has dramatically lowered the cost of financial transactions and expanded access to financial services.

Example: 20 years ago, paying a utility bill required a physical visit to a bank branch. Today, you can pay bills, transfer funds internationally, buy stocks, and apply for loans from your smartphone in seconds. Mobile banking apps, digital wallets (JazzCash, Easypaisa), and online brokerages are all forms of financial innovation.

4. Financial Crisis

A financial crisis occurs when the financial system stops working properly, causing sharp declines in asset prices, bank failures, and economic slowdown. It usually happens when borrowers cannot repay loans, leading to losses for banks and businesses. For example, during the 2007–2008 financial crisis, many people in the United States failed to repay home

loans, which caused major banks like Lehman Brothers to collapse, businesses to shut down, and unemployment to increase.

Structure of the financial system — key types:

Institution Type	Main Function	Examples
Commercial Banks	Accept deposits, make loans	HBL, MCB, Citibank
Investment Banks	Underwrite securities, M&A; advisory	Goldman Sachs, Morgan Stanley
Insurance Companies	Risk pooling, long-term investment	State Life, Jubilee Life
Mutual Funds	Pool small investor savings	NIT, JS Growth Fund
Pension Funds	Manage retirement savings	EOBI, Private pension schemes
Central Bank	Monetary policy, lender of last resort	State Bank of Pakistan, U.S. Fed

6. Why Study Money and Monetary Policy?

1. Money (Money Supply)

Anything that is generally accepted as payment for goods and services or in the repayment of debts. The money supply includes physical currency AND deposits in banks. Changes in the money supply have powerful effects on inflation, interest rates, employment, and overall economic output.

Example: In Pakistan, M2 (broad money supply) includes: currency notes in circulation + demand deposits (current accounts) + time deposits (savings/fixed accounts). When the State Bank of Pakistan prints more money, M2 increases — often leading to inflation if output does not grow proportionally.

2. Business Cycles

The recurring pattern of expansion (growth) and contraction (recession) in economic activity over time. During expansions, output and employment rise. During recessions (contractions),

output falls and unemployment rises. Changes in the money supply are closely linked to business cycle fluctuations.

Example: The U.S. experienced a severe recession in 1981–82 when restrictive monetary policy (high interest rates to fight inflation) caused unemployment to exceed 10%. As monetary policy eased and money supply grew, the economy recovered and unemployment fell to 5% by 1989 — a classic business cycle.

3. Monetary Policy

The management of the money supply and interest rates by the central bank to achieve macroeconomic objectives: price stability (low inflation), high employment, and sustainable economic growth. Central banks use tools like open market operations, reserve requirements, and policy interest rates.

Example: When the U.S. Federal Reserve raises its federal funds rate target by 0.25%, it makes borrowing more expensive throughout the economy → businesses invest less → consumers spend less → inflation pressure eases. Pakistan's State Bank uses the Policy Rate in the same way.

4. The Federal Reserve (Central Bank)

The central bank of the United States, established in 1913. It controls the money supply, sets key interest rates, supervises banks, and acts as lender of last resort. Understanding the Fed is essential because its decisions affect economies worldwide.

Example: When the Fed raised rates from 0.25% to 5.5% between 2022–2023 to fight inflation, borrowing costs rose globally — Pakistani bonds, emerging market currencies, and commodity prices were all affected. The Fed's decisions truly have global reach.

Money supply and economic variables — key relationships:

1. Money growth → Interest rates (short run): Faster money growth initially lowers interest rates (more money available to lend). This was clearly visible in U.S. data from the 1960s and 1970s when money growth and interest rate movements tracked closely.
2. Money growth → Inflation (long run): Sustained excessive money growth raises the price level. Milton Friedman's famous dictum: 'Inflation is always and everywhere a monetary phenomenon.'
3. Money decline → Recessions: Every major U.S. recession since the 20th century was preceded by a significant decline in the money growth rate.
4. Money supply and output: In the short run, faster money growth can stimulate output and reduce unemployment. But in the long run, it only raises prices — a key lesson of monetary economics.

Important Distinction — Monetary vs. Fiscal Policy:

Monetary policy = Central bank manages money supply and interest rates.

Fiscal policy = Government manages taxation and spending.

Both affect inflation and GDP, but through different mechanisms and with different time lags.

7. Money, Inflation, and Fiscal Policy

1. Inflation

A sustained, ongoing increase in the general (average) price level of goods and services in an economy. It erodes the purchasing power of money. The inflation rate is the percentage increase in the aggregate price level from one year to the next.

Example: In 1970, a can of Coca-Cola cost \$0.10 in the U.S. By 2023, it cost \$1.50+. This 15-fold price increase over 53 years reflects cumulative inflation. In Pakistan, the inflation rate exceeded 35% in 2023 — meaning goods worth Rs. 100 in early 2022 cost Rs. 135 by early 2023.

2. Money Supply and Inflation — The Evidence

Across countries and time periods, the data consistently shows that higher money supply growth leads to higher inflation. Countries that rapidly expand their money supply invariably experience high inflation. Nobel Prize winner Milton Friedman called inflation 'a monetary phenomenon' — meaning governments cause inflation by printing too much money.

Example: In the early 1990s, Belarus had money growth of ~1,000% per year and inflation of ~900%. Brazil had ~2,000% money growth and ~1,400% inflation. Meanwhile, the U.S. had money growth of ~3–4% per year and inflation of ~3%. The correlation is striking and consistent.

3. Fiscal Policy

Government decisions about how much to spend (government expenditure) and how much to collect (taxation). When government spending exceeds tax revenue, a budget deficit results. When revenue exceeds spending, there is a budget surplus. Fiscal policy interacts with monetary policy and affects interest rates, inflation, and growth.

Example: Pakistan's federal budget has historically run deficits. When the government borrows heavily to cover deficits, it competes with private borrowers for funds □ interest rates rise □ businesses and consumers borrow less □ slower economic activity. If the central bank 'monetizes' the deficit (prints money to lend to the government), inflation surges.

8. The Analytic Framework: How We Study?

Rather than memorizing isolated facts, Mishkin's textbook builds understanding through a unified analytic framework — a set of core economic tools applied consistently throughout every topic. Once you master these tools, you can analyze any financial market or policy situation.

The six core analytical tools:

1. Demand for Assets	Why do people hold bonds, stocks, money? What makes an asset attractive?	e.g.: At lower prices, more people want to buy bonds → downward sloping demand curve.
2. Supply & Demand Equilibrium	Markets 'clear' when supply equals demand, determining price and quantity.	e.g.: When bond demand rises (people want safety), bond prices rise and interest rates fall.
3. Profit Motive	Banks and financial firms act to maximize profit — this drives their behavior.	e.g.: Banks charge higher rates on risky loans because default risk threatens their profit.
4. Transaction Costs	Real-world friction costs of doing business affect financial decisions.	e.g.: High transaction costs explain why most people use banks rather than lending directly.
5. Asymmetric Information	When one party knows more than the other — creates adverse selection and moral hazard.	e.g.: Borrowers know their creditworthiness better than lenders → banks must screen carefully.
6. Aggregate Supply & Demand	Economy-wide supply and demand determines output and price level.	e.g.: Expansionary monetary policy shifts aggregate demand right → higher output AND prices.

Key Terms:

Security	A claim on the issuer's future income or assets; a financial instrument that can be bought and sold.
Bond	A debt security promising periodic interest payments and principal repayment at maturity.
Interest Rate	The cost of borrowing money; price paid for renting funds (% per year).
Common Stock	A share of ownership in a corporation; entitles holder to dividends and residual assets.

Foreign Exchange Market	Market where one currency is converted to another; determines exchange rates.
Foreign Exchange Rate	Price of one country's currency in terms of another country's currency.
Financial Markets	Markets transferring funds from savers (surplus units) to borrowers (deficit units).
Financial Intermediaries	Institutions (banks, funds) that borrow from savers and lend to borrowers.
Banks	Depository institutions accepting deposits and making loans; main financial intermediary type.
E-Finance	Delivery of financial services using electronic information technology.
Money (Money Supply)	Anything generally accepted as payment; includes currency + bank deposits.
Business Cycles	Recurring expansions and contractions in aggregate economic activity.
Recession	A period of declining aggregate output; defined as two consecutive quarters of negative GDP growth.
Aggregate Output	Total production of goods and services in an economy (related to GDP).
Unemployment Rate	Percentage of the labor force that is jobless and actively seeking employment.
Inflation	Sustained increase in the overall/average price level of goods and services.
Inflation Rate	Percentage increase in the aggregate price level from one period to the next.
Aggregate Price Level	The average price of goods and services in an economy (measured by CPI, etc.).
Monetary Policy	Central bank management of the money supply and interest rates.
Monetary Theory	The study of how money affects interest rates, output, and the price level.
Central Bank	Government institution that conducts monetary policy and supervises the banking system.
Federal Reserve (Fed)	The central bank of the United States, established in 1913.
Fiscal Policy	Government decisions about spending levels and tax rates.

Budget Deficit	Occurs when government expenditure exceeds tax revenue in a fiscal year.
Budget Surplus	Occurs when government tax revenue exceeds expenditure in a fiscal year.
Gross Domestic Product (GDP)	Total market value of all final goods and services produced in a country in a year.

CHAPTER 02

An Overview of the Financial System

Section 1: Function Of Financial Markets

1. What is a Financial Market?

A financial market is any place or system where buyers and sellers trade financial instruments such as stocks, bonds, currencies, and derivatives. The most important economic function of financial markets is to channel funds from people who have saved surplus money (lender-savers) to people who need money (borrower-spenders).

Example: Inez the Inventor has designed a robot but has no funds to produce it. Walter the Widower has extra savings. A financial market connects them — Walter lends to Inez, the robot gets produced, and the economy improves.

2. Who are Lender-Savers and Borrower-Spenders?

Lender-Savers (Surplus Units)	Borrower-Spenders (Deficit Units)
1. Households (main lenders)	1. Business Firms (main borrowers)
2. Business Firms (sometimes)	2. Federal Government (largest borrower)
3. State & Local Governments	3. Households (mortgages, car loans)
4. Foreign entities	4. Foreign entities

3. Two Routes for Funds to Flow

1. Route 1 — Direct Finance:

Borrowers borrow directly from lenders in financial markets by selling securities (also called financial instruments). Securities are assets for the buyer but liabilities (IOUs) for the issuer.

Example: General Motors needs funds for a new factory. It sells bonds directly to savers in the bond market. Savers lend money; GM gets funds. The bond is an asset for the saver, a liability (debt) for GM

2. Route 2 — Indirect Finance (via Financial Intermediaries):

A financial intermediary (e.g., a bank) stands between lenders and borrowers. The bank collects savings deposits and uses those funds to give loans to borrowers.

Example: You deposit \$1,000 in a bank (you are the lender-saver). The bank loans that money to a small business (the borrower-spender). The bank is the intermediary in the middle.

4. Why Are Financial Markets So Important?

Financial markets create economic efficiency by moving funds from those with no productive opportunities to those who have them. They also improve consumer welfare by allowing people to buy things when they need them rather than waiting until they save enough.

Example: You just started working and want to buy a house, but have little savings yet. A financial market lets you borrow (mortgage) now and repay over time — you enjoy the house while young and the lender earns interest. Both are better off!

Remember: When financial markets break down (financial crises — e.g., Mexico, East Asia, Argentina), severe economic hardship and even political instability can result.

Section 2: Structure Of Financial Markets

1. Debt Markets vs. Equity Markets

1. Debt Instruments

A debt instrument (bond or mortgage) is a contractual agreement where the borrower pays fixed amounts (interest + principal) at regular intervals until the maturity date.

Types by maturity:

Short-term: Maturity less than 1 year (e.g., Treasury Bills)

Intermediate-term: Maturity 1–10 years

Long-term: Maturity more than 10 years (e.g., 30-year bonds)

Example: You lend \$1,000 by buying a 10-year bond. The company pays you \$50 interest every year for 10 years, and returns your \$1,000 at the end. You are a debt holder, not an owner.

2. Equity Instruments (Stocks)

Equities (common stocks) are claims to share in a company's net income and assets. They pay dividends and have no maturity date. Owning stock means you own a portion of the firm and have voting rights.

Advantage of equity: You benefit directly from company growth.

Disadvantage of equity: Equity holders are residual claimants — the company must pay ALL debt holders before paying equity holders.

Example: Apple has 1 million shares outstanding. You own 1 share = you own 1/1,000,000 of Apple, get 1/1,000,000 of its profits as dividends, and can vote on company decisions.

Key Term: Debt Market Size (2002) = \$20 trillion | Equity Market Size (2002) = \$11 trillion. The debt market is LARGER than the stock market!

2. Primary Markets vs. Secondary Markets

1. Primary Market

A primary market is where new securities are sold for the first time. The issuing company receives the funds. An investment bank assists by underwriting — guaranteeing a price and selling to the public.

Example: Apple issues 1 million NEW shares for the first time (IPO). Investors buy these new shares. Apple receives the money. Goldman Sachs (investment bank) underwrites the deal.

2. Secondary Market

A secondary market is where previously issued (secondhand) securities are bought and sold. The issuing company does NOT receive any new funds here.

Secondary markets serve TWO key functions:

1. Make financial instruments more liquid (easier to convert to cash quickly).
2. Determine the price of securities, which affects the price the firm can charge in the primary market.

Example: You buy Apple shares on the NYSE from another investor. Apple gets nothing from this trade

— only the seller gets cash. NYSE, NASDAQ, and bond markets are secondary markets.

Remember: Books on financial markets focus on SECONDARY markets because they determine prices and are where most daily trading activity occurs.

3. Organized Exchanges vs. Over-the-Counter (OTC) Markets

Organized Exchanges	OTC Markets
Central physical/electronic location	Dealers at different locations connected by computer
Buyers & sellers (or their brokers) meet	Dealers stand ready to buy/sell at stated prices
Examples: NYSE, American Stock Exchange, Chicago Board of Trade	Examples: US Government bond market, NASDAQ (historically)
Most large corporations' stocks traded here	US Govt bond market (largest trading volume) is OTC

Key Term: Brokers = agents matching buyers with sellers. Dealers = buy/sell for their own account at stated prices.

4. Money Market vs. Capital Market

Money Market	Capital Market
Short-term debt instruments (maturity < 1 year)	Long-term debt and equity instruments (maturity ≥ 1 year)
LESS price fluctuation = SAFER	MORE price fluctuation = RISKIER
More liquid (widely traded)	Less liquid
Used by corporations & banks for temporary surplus funds	Used by insurance companies, pension funds (long-term investors)
Examples: T-Bills, Commercial Paper, CDs	Examples: Stocks, Long-term bonds, Mortgages
Example: A company has \$5 million sitting idle for 3 months. It invests in 3-month T-Bills (money market instrument) — safe, liquid, earns interest. A pension fund needing 30-year returns buys corporate bonds (capital market instrument).	

Section 3: Internationalization Of Financial Markets

1. Why Has Internationalization Grown?

Two main reasons:

- **Large increases** in savings pools in foreign countries (e.g., Japan)
- **Deregulation** of foreign financial markets, enabling them to expand American corporations now tap international capital markets. Foreign corporations also raise funds from Americans. Foreigners are major investors in the US.

2. Foreign Bonds

A foreign bond is sold in a foreign country and denominated in that country's currency.

Example: German automaker Porsche sells a bond in the United States denominated in US DOLLARS — this is a foreign bond. US railroads in the 1800s were financed by foreign bonds sold in Britain.

3. Eurobonds

A Eurobond is a bond denominated in a currency OTHER than that of the country in which it

is sold. Over 80% of new international bond issues are Eurobonds. The Eurobond market is now larger than the US corporate bond market.

Example: A bond denominated in US DOLLARS sold in LONDON = Eurobond. Note: Most Eurobonds are denominated in US dollars, NOT in euros. The 'Euro' prefix means 'outside home country' not the euro currency.

Remember: TRICK: Eurobond does NOT mean a bond in euros. It means a bond sold OUTSIDE its home currency's country. A USD bond sold in London is a Eurobond!

4. Eurocurrencies and Eurodollars

Eurocurrencies = foreign currencies deposited in banks OUTSIDE the home country.

Eurodollars = US dollars deposited in foreign banks outside the US (or in foreign branches of US banks). They earn interest like short-term Eurobonds. Over \$190 billion in Eurodollars outstanding.

Example: A Pakistani company deposits \$1 million in a London bank. Those dollars are 'Eurodollars.' American banks borrow these Eurodollar deposits to fund their lending — an important source of bank funds.

5. World Stock Markets

The US stock market used to be by far the largest in the world, but foreign markets have grown in importance. In the mid-1980s, Japan's stock market (Nikkei 225) at times exceeded the US market in value. Investors now follow global indexes:

- Nikkei 225 Average — Tokyo, Japan
- FTSE 100-Share Index — London, UK
- Dow Jones Industrial Average — New York, USA

Section 4: Function Of Financial Intermediaries

1. What is a Financial Intermediary?

A financial intermediary stands between lender-savers and borrower-spenders. It borrows from lender-savers and lends to borrower-spenders. This process is called financial intermediation — the PRIMARY route for moving funds.

Financial intermediaries are a far more important source of financing for corporations than securities markets — true worldwide, not just in the US!

Example: Bank (Honest John Bank): Collects deposits from you (lender-saver), lends those funds to General Motors (borrower-spender). The bank is the intermediary.

2. Reason 1 — Transaction Costs

Transaction costs = time and money spent carrying out financial transactions. They are a major problem for direct lending. Financial intermediaries reduce transaction costs through economies of scale — as transaction size increases, cost per transaction falls.

Economies of Scale: Large size allows fixed costs to be spread over more transactions.

Example: Problem: You want to lend Carl the Carpenter \$1,000 but need a lawyer to write a contract = costs \$500. You can't profit from just \$100 interest. Solution: A bank hires a lawyer for \$5,000 to write ONE perfect contract used for 2,000 loans = only \$2.50 per loan. Now it's profitable!

Low transaction costs also allow intermediaries to provide liquidity services — making it easy to conduct transactions (e.g., checking accounts, ATMs, online banking).

3. Reason 2 — Risk Sharing (Asset Transformation)

Risk = uncertainty about the returns you will earn. Financial intermediaries help reduce investor risk through **risk sharing**: they create and sell low-risk assets to savers, then use those funds to purchase higher-risk assets.

This is also called **asset transformation** — risky assets are transformed into safer assets for investors.

Diversification: Spreading investments across many assets (portfolio) so that overall risk is lower. ('Don't put all eggs in one basket!')

Example: You deposit \$1,000 in a bank savings account (safe, low-risk). The bank pools funds from thousands of depositors and lends to many different borrowers. Even if some loans default, the overall risk is spread out. Your deposit remains safe.

4. Reason 3 — Asymmetric Information

Asymmetric information = one party in a transaction knows more than the other. In finance, borrowers usually know more about their investment risk than lenders. This creates two major problems:

1. Adverse Selection (BEFORE the transaction)

Adverse selection = the problem that the most likely borrowers to seek loans are those who are BAD credit risks. Good risks don't need to borrow as desperately. This may cause lenders to refuse ALL loans.

Example: Two aunts: Aunt Louise (conservative, will repay) and Aunt Sheila (gambler, may not repay). Which aunt calls you more often for a loan? AUNT SHEILA! She has more to gain. Because you can't tell them apart, you might lend to neither — this hurts even Aunt Louise who is a good risk. This is adverse selection.

Remember: MEMORY TIP: Adverse Selection = BEFORE the deal. Bad people SELECT themselves into the pool.

2. Moral Hazard (AFTER the transaction)

Moral hazard = after you lend money, the borrower may engage in riskier activities than agreed, making repayment less likely. Since you can't monitor them perfectly, they may misbehave.

Example: You lend Uncle Melvin \$1,000 for a word processor business. After getting the money, Uncle Melvin goes to the racetrack to gamble instead! If he wins big, he pockets the profit. If he loses, you bear the cost. He has an incentive to take risks with YOUR money.

Remember: MEMORY TIP: Moral Hazard = AFTER the deal. Borrower behaves in a 'morally hazardous' way.

3. Economies of Scope and Conflicts of Interest

Financial intermediaries play an important role by providing multiple financial services such as loans, investment advice, and selling bonds. By offering many services together, they achieve economies of scope, which means they reduce costs by using the same information for different services.

For example, when a bank evaluates a company for a loan, it can use the same information to help sell that company's bonds. However, this can create conflicts of interest, where financial institutions may favor their own benefits and provide misleading information. For example, a bank may recommend investing in a company whose bonds it is selling, even if the investment is risky. This reduces trust, creates information problems, and makes financial markets less efficient.

5. How Intermediaries Solve These Problems

Against Adverse Selection: Financial intermediaries are better at screening good from bad credit risks than individual lenders. They have expertise and information.

Against Moral Hazard: Intermediaries monitor borrowers after lending, reducing the risk of misbehavior.

Example: A bank checks your credit history, income, and collateral before giving a loan (screening). Then it requires monthly statements and may inspect the business (monitoring). An individual lender has neither the tools nor time to do this.

Section 5: Types Of Financial Intermediaries

Financial intermediaries fall into THREE main categories:

Category A: Depository Institutions (Banks)

These accept deposits from individuals/institutions and make loans. They are involved in creating deposits, an important part of the money supply.

1. Commercial Banks

The largest and most diversified intermediary (~8,000 in the US). Raise funds via checkable deposits, savings deposits, and time deposits. Use funds for commercial, consumer & mortgage loans, and to buy government securities.

Example: Citibank, JP Morgan Chase — you deposit your salary; the bank lends it to businesses

2. Savings & Loan Associations (S&Ls) and Mutual Savings Banks

~1,500 institutions. Obtain funds through savings/time deposits. Historically focused on residential mortgage loans. Restrictions have been loosened; now more similar to commercial

banks.

Example: Washington Mutual was an S&L.; You save; they give you a home mortgage loan.

3. Credit Unions

~9,500 small cooperative institutions organized around a specific group (union members, company employees). Raise funds from 'shares' (deposits). Make mainly consumer loans.

Example: A teachers' credit union: Only teachers can join. They pool savings and lend to each other at lower rates.

Acquire funds periodically on a contractual basis. Can predict cash outflows accurately, so they invest in long-term securities (less need for liquidity).

Category B: Contractual Savings Institutions

1. Life Insurance Companies

Insure against financial hazards after death; sell annuities. Collect premiums from policyholders. Invest mainly in corporate bonds and mortgages. Assets = \$3.3 trillion (largest contractual savings institution).

Example: MetLife, Prudential — you pay a monthly premium; they invest it; they pay your family if you die.

2. Fire and Casualty Insurance Companies

Insure against theft, fire, accidents. Receive premiums. Because they face potential large sudden losses, they hold more liquid assets than life insurers. Largest holding = municipal bonds.

Example: State Farm insures your house. If a fire occurs, they pay. They hold liquid assets to pay claims quickly.

3. Pension Funds & Government Retirement Funds

Provide retirement income (annuities). Funds come from employer/employee contributions. Largest asset holdings = corporate bonds and stocks. Encouraged by government through tax incentives and legislation.

Example: Your employer deducts pension contributions from your salary. These funds are invested in stocks and bonds and given back to you at retirement.

Category C: Investment Intermediaries

1. Finance Companies

Raise funds by selling commercial paper, stocks, and bonds. Lend to consumers (furniture,

cars, home improvements) and small businesses. Some are subsidiaries of manufacturing companies.

Example: Ford Motor Credit Company: Ford's finance arm gives you a loan to buy a Ford car.

2. Mutual Funds

Raise funds by selling shares to many individuals; use proceeds to buy diversified portfolios of stocks/bonds. Allow shareholders to pool resources for lower transaction costs. Shares can be redeemed anytime but values fluctuate (risky).

Example: Fidelity Magellan Fund: You buy a \$100 share; Fidelity pools everyone's money and buys a diversified portfolio of 100 stocks — reducing your risk.

3. Money Market Mutual Funds

Like mutual funds but also function like banks — offer deposit-type accounts. Invest in safe, liquid money market instruments. Key feature: shareholders can write checks against their shares (like a checking account that pays interest). Assets grew to nearly \$2.1 trillion by 2002.

Example: Fidelity Cash Reserves: You put in \$5,000; it earns interest; you can write checks against it. Better than a regular bank account in returns, but slightly less regulated.

Summary Table — Intermediaries

Type	Main Source of Funds (Liabilities)	Main Use of Funds (Assets)
Commercial Banks	Deposits	Business, consumer, mortgage loans; govt securities
S&Ls / Mutual Savings	Deposits	Mortgages
Credit Unions	Shares (deposits)	Consumer loans
Life Insurance	Premiums	Corporate bonds, mortgages

Fire/Casualty Insurance	Premiums	Municipal bonds, corporate bonds, stocks
Pension Funds	Employer/employee contributions	Corporate bonds and stocks
Finance Companies	Commercial paper, bonds, stocks,	Consumer and business loans
Mutual Funds	Shares	Stocks, bonds
Money Market Mutual Funds	Shares	Money market instruments

Section 6: Regulation Of The Financial System

1. Two Main Reasons for Government Regulation

- Increase information available to investors (combat adverse selection & moral hazard)
- Ensure the soundness of the financial system (prevent financial panics)

2. Increasing Information — The SEC

The Securities and Exchange Commission (SEC) was established in 1933 after the 1929 stock market crash and widespread fraud. It requires corporations to disclose sales, assets, and earnings publicly, and restricts insider trading.

Example: Before buying Apple stock, you can look up Apple's annual report (10-K filing) on the SEC's EDGAR database. Apple must disclose its revenues, debts, and risks by law.

➤ Ensuring Soundness — Six Types of Regulation

1. Restrictions on Entry

Only upstanding citizens with strong credentials and capital can get a banking/insurance charter from state or federal government.

Example: You can't just open a bank. You need a charter, minimum capital, regulatory approval — to prevent fraudsters from running banks.

2. Disclosure Requirements

Financial intermediaries must follow strict bookkeeping standards, allow periodic inspection, and make information public.

Example: Banks must publish quarterly financial statements so depositors can check their health.

3. Restrictions on Assets and Activities

Limits on what intermediaries can do and hold. Banks cannot hold common stocks (too risky). Insurance companies can hold some but with limits.

Example: Glass-Steagall Act (1933, repealed 1999): Separated commercial banking from investment banking to prevent risky speculation.

4. Deposit Insurance

FDIC insures each depositor up to \$100,000 per account. Prevents bank runs — depositors don't panic because deposits are protected.

Example: FDIC was created in 1934 after massive bank failures of 1930-1933. Similar agencies: NCUSIF for credit unions, SAIF for S&Ls.;

5. Limits on Competition

Restrictions on bank branching historically prevented banks from opening too many branches (seen as reducing failure risk, though evidence is weak).

Example: Branching restrictions were abolished by federal law in 1994.

6. Restrictions on Interest Rates

After 1933, banks were prohibited from paying interest on checking accounts. 'Regulation Q' set maximum interest on savings accounts (until 1986).

Example: These regulations were thought to prevent bank failures during competition, but later evidence doesn't support this, so they were abolished.

➤ Principal Regulatory Agencies

Agency	Regulates	Key Function
SEC	Organized exchanges & financial markets	Requires disclosure; restricts insider trading
CFTC	Futures market exchanges	Regulates futures trading procedures
Comptroller of Currency	Federally chartered commercial banks	Charters, examines books, restricts assets
NCUA	Federally chartered credit unions	Charters, examines, restricts assets

FDIC	Commercial banks, mutual savings banks, S&Ls	Insures deposits up to \$100,000; examines banks
Federal Reserve	All depository institutions	Examines commercial banks; sets reserve requirements
Office of Thrift Supervision	Savings and loan associations	Examines S&Ls; restricts assets

Section 7: Financial Market Instruments

1. Money Market Instruments (Short-term, Maturity < 1 Year)

These are SAFEST and MOST LIQUID instruments due to short maturities and less price fluctuation.

1. US Treasury Bills (T-Bills)

Short-term debt of the US government. Issued in 3-, 6-, 12-month maturities. Pay NO explicit interest — instead sold at a DISCOUNT (below face value). MOST LIQUID of all money market instruments. Almost ZERO default risk.

Example: You pay \$9,000 today for a T-Bill that pays \$10,000 in 12 months. Your interest = \$1,000.

2. Negotiable Bank CDs (Certificates of Deposit)

Debt instrument sold by a bank. Pays annual interest; at maturity returns principal. Citibank introduced negotiable CDs in 1961 (can be resold in secondary market). Now ~\$1.2 trillion outstanding. Held by corporations, money market mutual funds.

Example: You put \$100,000 in a 6-month CD at 4% interest. At maturity you get \$102,000 back.

3. Commercial Paper

Short-term debt issued by large banks and well-known corporations. Growth: 3,900% from 1970-2002 (from \$33 billion to \$1.3 trillion). Corporations use it instead of bank loans for short-term borrowing — cheaper direct finance.

Example: AT&T; needs \$50 million for 90 days. It issues commercial paper to money market funds instead of borrowing from a bank — saves on interest.

4. Banker's Acceptances

Bank draft (like a check) issued by a firm, guaranteed by a bank for a fee. Used mainly in

INTERNATIONAL TRADE. The bank guarantee reassures foreign exporters. Can be resold at a discount. Held mainly by same parties as T-Bills.

Example: A Pakistani company imports goods from Japan. To assure the Japanese exporter of payment, Habib Bank 'accepts' (guarantees) the draft — creating a banker's acceptance.

5. Repurchase Agreements (Repos)

Short-term loans (usually < 2 weeks) using T-Bills as collateral. Introduced in 1969. Now over \$400 billion outstanding. Important source of bank funds. Main lenders = large corporations.

Example: GM has \$1 million idle for one week. It 'buys' T-Bills from a bank, which agrees to buy them back in a week at a slightly higher price. Effectively a 1-week loan from GM to the bank, secured by T-Bills.

6. Federal Funds

Overnight loans BETWEEN BANKS of their deposits at the Federal Reserve. NOT loans from the federal government. Banks borrow when they don't have enough reserves. The FEDERAL FUNDS RATE is a key indicator of monetary policy tightness.

Example: Bank A is short on reserves by \$5 million. It borrows overnight from Bank B through the Fed's wire system at the federal funds rate.

7. Eurodollars

US dollars deposited in foreign banks outside the US. Short-term (like short-term Eurobonds). Over \$190 billion outstanding. Important source of funds for American banks.

Example: A Saudi company deposits \$500 million in a London bank. Those are Eurodollars. A US bank borrows those Eurodollar deposits to fund domestic loans.

2. Capital Market Instruments (Long-term, Maturity $\geq$ 1 Year)

These have WIDER price fluctuations and are RISKIER, but offer higher returns.

1. Stocks (Corporate Equities)

Equity claims on net income and assets. Total value = \$11 trillion (end 2002). New stock issues each year = less than 1% of total. About half held by individuals; rest by pension funds, mutual funds, insurance companies.

Example: You own 100 shares of Apple. If Apple earns \$1 billion profit and has 1 billion shares, you receive \$100 as dividends. If the stock price rises from \$100 to \$150, your investment gains \$5,000.

2. Mortgages

Loans to households/firms to buy housing, land, or structures. LARGEST DEBT MARKET in the US. Residential mortgages = 4x commercial mortgages. S&Ls and mutual savings banks are primary lenders for residential mortgages. Govt agencies: Fannie Mae, Ginnie Mae, Freddie Mac buy mortgages and provide liquidity.

Example: You borrow \$200,000 from a bank to buy a house. The house is collateral. The bank may sell your mortgage to Fannie Mae, which bundles it into a mortgage-backed security and sells to investors.

3. Corporate Bonds

Long-term bonds issued by corporations with strong credit. Pays interest twice a year, returns face value at maturity. Some are CONVERTIBLE BONDS (can convert to stock). Corporate bond market is smaller than stock market but new issues are much larger annually. Main buyers = life insurance companies.

Example: You borrow \$200,000 from a bank to buy a house. The house is collateral. The bank may sell your mortgage to Fannie Mae, which bundles it into a mortgage-backed security and sells to investors.

4. US Government Securities

Long-term debt issued by the US Treasury to finance budget deficits. MOST WIDELY TRADED bonds in the US (over \$100 billion daily volume). Most LIQUID capital market security. Held by: Federal Reserve, banks, households, foreigners.

Example: The US government borrows \$1 trillion by issuing 10-year Treasury Bonds at 4%. China and Japan hold massive amounts of these bonds.

5. US Government Agency Securities

Long-term bonds issued by govt agencies (Ginnie Mae, Freddie Mac, Federal Farm Credit Bank, TVA) to finance mortgages, farm loans, power equipment. Many are guaranteed by the federal government.

Example: Ginnie Mae issues bonds and uses proceeds to buy mortgages from banks, providing more funds for housing loans.

6. State and Local Government Bonds (Municipal Bonds)

Long-term debt of state & local governments. Fund schools, roads, large programs. Key feature: interest payments are EXEMPT from federal income tax (and often state tax). Main buyers = commercial banks (due to high tax rates) and wealthy individuals.

Example: The city of Karachi issues a 20-year municipal bond at 3% tax-free to fund a new highway. A high-income investor in the 35% tax bracket earns 3% tax-free —

equivalent to ~4.6% taxable!

7. Consumer and Bank Commercial Loans

Loans to consumers and businesses made by banks and finance companies. Often NO secondary markets — least liquid of all capital market instruments. Secondary markets are developing rapidly.

Example: Your car loan from a bank, or a small business line of credit — these are commercial and consumer loans held on bank balance sheets.

The 6 Key Points of the Chapter 2

Point 1	Financial markets channel funds from lender-savers to borrower-spenders via DIRECT FINANCE (securities) or INDIRECT FINANCE (intermediaries). This increases economic efficiency.
Point 2	Financial markets are classified as: Debt vs Equity Primary vs Secondary Exchanges vs OTC Money vs Capital
Point 3	International trend: Eurobonds now dominate the international bond market (>80% of new issues). Eurodollars are a major funding source for US banks.
Point 4	Financial intermediaries reduce TRANSACTION COSTS, allow RISK SHARING, and solve ADVERSE SELECTION and MORAL HAZARD problems.
Point 5	Three types of intermediaries: (a) Banks — commercial, S&Ls, credit unions (b) Contractual savings — life/casualty insurance, pension funds (c) Investment — finance companies, mutual funds, money market funds
Point 6	Government regulates for two reasons: (1) Increase investor information (2) Ensure soundness via entry restrictions, disclosure, asset restrictions, deposit insurance, competition limits, and interest rate caps.

CHAPTER 03

What is Money?

1. Meaning of Money

1. What is Money?

In everyday life, people use the word 'money' loosely — but in economics, it has a very precise meaning. Money (or money supply) is defined as anything that is generally accepted as payment for goods or services, or in the repayment of debts.

Example: Currency (paper bills and coins) is money. But not everything valuable is money.

Money vs. Currency vs. Wealth vs. Income

Term	Definition	Example
Money	Anything accepted as payment for goods/services or repayment of debts	Cash, checking account
Currency	Paper bills and coins only (too narrow for economists)	A \$10 bill
Wealth	Total collection of property that stores value	Bonds, cars, houses, stocks, art
Income	A FLOW of earnings per unit of time	\$5,000/month salary

Key Point: Money is a STOCK (fixed amount at a point in time). Income is a FLOW (over time). If someone has an income of \$1,000, you don't know if it's per day, month, or year. But if they have \$1,000 in their pocket — that's exact!

2. Functions of Money

Money serves **THREE primary functions** in any economy — whether money is shells, rocks, gold, or paper:

1. Medium of Exchange

Money is used as a medium of exchange — it is the thing people give in return for goods and services. This is money's MOST important function and what distinguishes it from other assets like stocks or bonds.

Without money, we would have a barter economy — where goods are exchanged directly for other goods. This creates the problem of 'double coincidence of wants': you must find someone who has what you want AND wants what you have.

Example — Ellen the Economics Professor: In a barter economy, if Ellen wants food, she must find a farmer who not only produces food she likes but also wants to learn economics from her. This search is time-consuming and may never succeed — she might starve! With money, Ellen teaches anyone, gets paid, then buys food from any farmer. Problem solved!

The time/effort spent searching in barter is called a transaction cost. Money eliminates this by being universally accepted, encouraging specialization and division of labor.

For a commodity to work as money, it must be:

- Easily standardized (value can be ascertained simply)
- Widely accepted (everyone must take it)
- Divisible (easy to 'make change')
- Easy to carry (portable)
- Not deteriorate quickly (durable)

Historical Examples: Wampum (beads) by Native Americans; tobacco & whiskey by early American colonists; cigarettes in WWII prisoner-of-war camps — all served as money!

2. Unit of Account

Money acts as a unit of account — a standard measurement for stating the prices of goods and services. Just like weight is measured in pounds and distance in miles, value is measured in dollars.

Why it matters — The Price Problem: In a barter economy with 1,000 goods, you would need 499,500 different prices (one for each pair of goods). With money, you only need 1,000 prices — one per good in dollars. This massively reduces transaction costs!

Formula for number of prices needed in barter: $N(N-1)/2$. With 10 goods = 45 prices. With 100 goods = 4,950 prices. With 1,000 goods = 499,500 prices!

3. Store of Value

Money also functions as a store of value — a way to save purchasing power from when income is received until when it is spent. It allows people to shop later rather than immediately.

Money is NOT unique as a store of value — stocks, bonds, land, art, houses also store value. In fact, those often store value BETTER (they pay interest or appreciate). So why hold money?

Answer: LIQUIDITY — the relative ease and speed with which an asset can be converted into a medium of exchange. Money is the most liquid asset. Selling a house takes time and costs 4-6% commission. Money needs no conversion.

Example: If you urgently need cash, your house is useless unless you can sell it quickly — and rushing means accepting a lower price. A \$100 bill? Instantly usable.

Inflation Problem: During high inflation, money loses its value rapidly as a store of value. In Germany after WWI (hyperinflation of 1923), the price level rose over 30 billion times. Workers were paid daily so they could spend wages before they lost value by evening!

Quick Summary: Functions of Money

Function	What it means	Benefit
Medium of Exchange	Used to pay for goods/services	Eliminates barter, reduces transaction costs
Unit of Account	Standard measure of value	Reduces number of prices needed
Store of Value	Saves purchasing power over time	Enables future purchases; most liquid

3. Evolution of the Payment Systems

The **payments system** is the method of conducting transactions in an economy. It has been evolving over centuries. Understanding this evolution helps us understand what money might look like in the future.

1. Commodity Money

Commodity money is money made up of precious metals (gold, silver) or another valuable commodity. It was the main form of money from ancient times until several hundred years ago.

Example: Gold coins were used throughout the ancient world. Everyone valued gold, so it was universally accepted. Problem: Gold is HEAVY. Buying a house with gold coins would require renting a truck to transport the payment!

Disadvantage: Precious metals are very heavy and hard to transport in large amounts.

2. Fiat Money

Paper currency evolved from receipts for gold deposits. Over time it became fiat money — paper currency decreed by governments as legal tender (must be accepted for payment of debts), but NOT convertible into coins or precious metal.

Example: The US dollar today. You cannot walk into the Federal Reserve and exchange your \$20 bill for gold. It's valuable because the government says so, and because we all trust and accept it.

Fiat money works only when: (1) There is some trust in the authorities, and (2) Counterfeiting is very difficult. European countries replaced their currencies with the euro in 2002 — a legal arrangement, not commodity-based.

Disadvantage: Paper currency and coins can be stolen and are expensive to transport in large amounts.

3. Checks

A **check** is an instruction to your bank to transfer money from your account to someone else's account. This eliminated the need to carry large amounts of cash.

Example: Paying rent of \$50,000 with cash would require lugging a heavy bag. A check transfers the same amount with a piece of paper — much safer and easier.

Advantages of Checks:

- No need to carry large amounts of currency
- Payments cancel each other — reduced physical currency movement
- Can be written for any exact amount
- Reduced theft risk
- Provides a receipt/record of purchase

Disadvantages of Checks:

- Takes time for check to travel between locations
- Several business days before funds are available
- Processing cost estimated at over \$10 billion/year in the US

4. Electronic Payment

Cheap computers and the Internet made it possible to pay bills electronically — no stamps, no checks, no delays.

Example: Online bill pay — you log on to your bank's website, click a few buttons, and your electricity bill is paid instantly. Recurring bills can be set to auto-deduct. Saves over \$1 per transaction vs. writing a check.

5. E-Money (Electronic Money)

E-money is money that exists only in electronic form. It substitutes for both checks AND cash.

Type	How it works	Example
Debit Card	Electronically transfers funds from your bank account to merchant's account	Swiping a Visa debit card at a supermarket
Stored-Value Card	Pre-paid card loaded with a fixed dollar amount	Prepaid phone card
Smart Card	Contains a computer chip; can be loaded from your bank via ATM/PC	Transit cards in Japan/Korea
E-Cash	Used on the Internet to buy goods; transferred computer-to-computer	Online purchases via a digital wallet

FYI: Are We Headed for a Cashless Society?

Despite the convenience of e-money, a fully cashless society has not arrived. Several factors keep paper money alive:

- It is expensive to set up computers, card readers, and telecom networks
- Electronic payments raise security and privacy concerns (hackers, data theft)
- Electronic trails contain personal buying-habit data (privacy issue)
- E-money pilot programs (like Mondex smart cards in Britain, 1995) have mostly failed

Conclusion: As paraphrased from Mark Twain — 'the reports of cash's death are greatly exaggerated.'

4. Application: Will Bitcoin Become The Money Of The Future?

Bitcoin is a new type of electronic money created in 2009 by software developer Satoshi Nakamoto. It is NOT controlled by any central bank. New bitcoins are created in a **decentralized** way by users who verify and process transactions using computing power — a process called '**mining**.'

Function of Money	Bitcoin's Performance	Verdict
Medium of Exchange	Transaction fees lower than credit cards; transactions can be anonymous	Works well
Unit of Account	Almost no one prices products in bitcoins; too volatile	Fails
Store of Value	Extreme price swings (e.g. 30 cents to \$32, then \$1125, then below \$500); 7x more volatile than gold	Fails (too risky)

Conclusion: Bitcoin fails 2 of the 3 key functions of money. Governments worry about its use in drug transactions and money laundering. China has outlawed it. Mt. Gox suffered ~\$500 million in Bitcoin theft (2014). Bitcoin is unlikely to be the money of the future — but its underlying technology (cheap electronic transactions) may become a feature of future payment systems.

5. MEASURING MONEY

Since money is defined by behavior (what people accept as payment), measuring it precisely is difficult. The **Federal Reserve System (the Fed)** — the central bank of the US — uses precise definitions called **monetary aggregates** to measure the money supply.

1. M1 — The Narrowest Measure

M1 includes the MOST LIQUID assets only:

Interesting Fact (FYI Box, Page 104): More than \$4,000 in US currency exists per person in America — but most people don't carry that much! Who has it? Criminals (prefer cash, untraceable), and Foreigners (people in countries with high inflation hoard US dollars). Over half of all US dollars are held abroad!

Component	Value (Aug 2014)	What it includes
Currency	\$1,206.1 billion	Paper money & coins held by the nonbank public (NOT cash in ATMs or bank vaults)
+ Traveler's Checks	\$3.3 billion	Only traveler's checks NOT issued by banks
+ Demand Deposits	\$1,089.9 billion	Business checking accounts that pay NO interest (also includes bank-issued traveler's checks)
+ Other Checkable Deposits	\$477.4 billion	Interest-bearing checking accounts (NOW accounts, ATS accounts) held by households
= TOTAL M1	\$2,776.7 billion	All directly spendable assets

Historical Note: Until the mid-1970s, only commercial banks could have checking accounts, and they paid no interest. Financial innovation changed this — savings banks, mutual savings banks, and credit unions can now offer checking accounts too.

2. M2 — The Broader Measure

M2 includes everything in M1 PLUS other assets that are less liquid but can be quickly converted to cash:

Component	Value (Aug 2014)	What it includes
M1	\$2,776.7 billion	All M1 components above
+ Small-Denomination Time Deposits	\$533.0 billion	CDs under \$100,000; redeemed without penalty only at fixed maturity date
+ Savings Deposits & Money Market Deposit Accounts	\$7,338.2 billion	Non-transaction deposits added/withdrawn any time
+ Money Market Mutual Fund Shares (retail)	\$642.5 billion	Retail accounts on which households can write checks; issued by banks

= TOTAL M2	\$11,290.4 billion	All M1 + near-money assets
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Example: A savings account at a bank — you can't write checks directly from it, but you can transfer money to your checking account any time and then spend it. So it's 'near money' but not directly spendable.

3. M1 vs M2 — Why Both Matter

Economists and policymakers are not sure which is the BEST measure of money. If M1 and M2 move together, it doesn't matter which one they use. But if they move in OPPOSITE directions, it creates confusing and conflicting policy recommendations.

Evidence from Figure 1 (Growth Rates 1960–2014): Generally, M1 and M2 grow together. BUT in 1992–1994 and 2004–2007, they moved in opposite directions. In 2009–2011, M1 surged over 15% while M2 rose much less dramatically — telling very different stories about monetary policy.

Key Takeaway: Choosing the right monetary aggregate DOES matter. The data in Figure 1 shows that using M1 vs M2 can lead policymakers to completely different conclusions about the money supply and what actions to take.

Summary

1. Money $\neq$ Income $\neq$ Wealth	Money = anything accepted as payment for goods/services or repayment of debts. Income = flow of earnings over time. Wealth = total stock of assets.
2. Three Functions of Money	Medium of Exchange (eliminates barter & double coincidence of wants) Unit of Account (standard for measuring value) Store of Value (saves purchasing power; money is most liquid).
3. Payments System Evolution	Commodity Money ☐ Fiat Money ☐ Checks ☐ Electronic Payment ☐ E-Money. Each stage solved a problem of the previous stage.
4. Liquidity	The ease of converting an asset into a medium of exchange. Money = most liquid. House = least liquid. Liquidity is why we hold money despite lower returns.
5. Hyperinflation destroys money's store of value	Germany 1923: prices rose 30 billion times in 2 years. People spent wages instantly before they lost value.
6. Bitcoin	Works as medium of exchange but fails as unit of account and store of value due to extreme price volatility. NOT the money of the future.
7. M1 (Narrow Money)	Currency + Traveler's Checks + Demand Deposits + Other Checkable Deposits = \$2,776.7B (Aug 2014). Most liquid assets.
8. M2 (Broad Money)	M1 + Small Time Deposits + Savings Deposits + Money Market funds = \$11,290.4B (Aug 2014). Less liquid but still near-money.
9. The Fed publishes M1 & M2 weekly	Released every Thursday in the H.6 release. Movements of M1 and M2 can diverge, making monetary policy decisions complex.

Key Terms

Money (Money Supply)	Anything generally accepted as payment for goods/services or repayment of debts
Currency	Paper bills and coins
Wealth	Total collection of property that stores value (bonds, stocks, land, cars, houses, etc.)
Income	A flow of earnings per unit of time
Medium of Exchange	Anything used to pay for goods and services

Barter Economy	Economy where goods are exchanged directly for other goods (no money)
Transaction Cost	Time and resources spent in exchange (high in barter, low with money)
Double Coincidence of Wants	The problem in barter: both parties must want what the other has
Unit of Account	A standard used to measure value in an economy
Store of Value	A repository of purchasing power available over time
Liquidity	Ease and speed with which an asset is converted into a medium of exchange
Hyperinflation	Extremely high inflation (exceeding 50% per month)
Payments System	Method of conducting transactions in the economy
Commodity Money	Money made of precious metals or another valuable commodity
Fiat Money	Paper currency declared legal tender by government; not convertible to precious metals
E-Money	Money that exists only in electronic form
Debit Card	Card that electronically transfers funds from bank account to merchant
Smart Card	Stored-value card with a computer chip that can be loaded from a bank account
E-Cash	Electronic cash used on the Internet to purchase goods
Monetary Aggregates	Official measures of the money supply (M1, M2) published by the Federal Reserve
M1	Narrowest money measure: currency + traveler's checks + demand deposits + other checkable deposits
M2	Broader measure: M1 + small time deposits + savings deposits + money market fund shares
Demand Deposits	Business checking accounts that do not pay interest
Small-Denomination Time Deposits	CDs under \$100,000; redeemable without penalty only at maturity

CHAPTER 04

The Meaning of Interest Rates

1. Introduction – Why Interest Rates Matter

1. What are Interest Rates?

Interest rates are one of the most closely watched variables in economics. They affect virtually every financial decision we make – from saving and borrowing to buying a house or investing in a business. The chapter's primary goal is to define interest rates precisely and explain how they are measured.

Real-Life Impact

1. **Personal Decisions:** Whether to save or spend, buy a house, or invest in bonds.
2. **Business Decisions:** Whether to invest in new equipment or hold cash in a bank.
3. **Economy-wide:** Interest rate movements are reported daily by news media because of their broad impact.

2. What This Chapter Covers

1. The Yield to Maturity (YTM) – the most accurate measure of interest rates.
2. How to calculate YTM for four types of debt instruments.
3. Alternative (less accurate) measures: Current Yield and Discount Yield.
4. Why a bond's interest rate $\neq$ its rate of return.
5. Difference between Real and Nominal interest rates.

2. Present Value (PV) – The Core Concept

1. What is Present Value?

Present Value (PV) is based on one simple idea: a dollar today is worth MORE than a dollar in the future because money today can be invested to earn interest.

■ Insight

Imagine you deposit \$100 in a bank at 10% interest.

After 1 year $\rightarrow \$100 \times (1 + 0.10) = \mathbf{\$110}$

After 2 years $\rightarrow \$100 \times (1.10)^2 = \mathbf{\$121}$

After 3 years $\rightarrow \$100 \times (1.10)^3 = \mathbf{\$133}$

So, \$100 TODAY = \$133 THREE YEARS FROM NOW (at 10% interest).

2. The Present Value Formula

$$PV = FV \div (1 + i)^n$$

Symbol	Meaning
PV	Present Value – today's value
FV	Future Value – amount received in the future
i	Interest rate (decimal form, e.g. 10% = 0.10)
n	Number of years into the future

■ Example – The \$20 Million Lottery

You win a lottery promising \$1 million per year for 20 years (total = \$20 million). But in **present value terms** at 10% interest, it is worth much less!

Year 1 payment: $\$1,000,000 \div (1.10)^1 = \mathbf{\$909,090}$

Year 2 payment: $\$1,000,000 \div (1.10)^2 = \mathbf{\$826,446}$

... and so on. Adding all 20 payments = **~\$9.4 million** in today's dollars.

Key Lesson: You didn't really win \$20 million – you won about half that amount in real terms!

3. Four Types Of Credit Market Instruments

Different debt instruments make payments at different times. Understanding each type is essential for calculating interest rates correctly.

Type 1: Simple Loan

The lender gives the borrower a principal amount. At maturity, the borrower repays the principal PLUS an interest payment. Most commercial loans are of this type.

■ Example

You lend your friend Jane \$100 for 1 year at 10% interest.

Jane repays \$100 (principal) + \$10 (interest) = **\$110** after 1 year.

Simple interest rate $i = \$10 \div \$100 = \mathbf{10\%}$

Type 2: Fixed-Payment Loan (Fully Amortized Loan)

The borrower repays the loan in equal periodic payments that include both principal AND interest. Common examples: car loans, mortgages.

$$LV = FP/(1+i) + FP/(1+i)^2 + \dots + FP/(1+i)^n$$

■ Example

You borrow \$1,000. You repay **\$126 per year for 25 years**.

YTM can be found using a financial calculator → Answer: **i = 12%**

Each payment covers part interest + part principal until fully paid off.

Type 3: Coupon Bond

Pays a fixed coupon (interest) payment every year AND returns the face value at maturity. Identified by: Issuer, Maturity Date, and Coupon Rate.

Key Terms:

1. **Face Value (Par Value):** Amount repaid at maturity (usually \$1,000).
2. **Coupon Payment:** Fixed annual interest payment (e.g. \$100).
3. **Coupon Rate:** Coupon Payment ÷ Face Value × 100 (e.g. \$100/\$1,000 = 10%).

Examples: U.S. Treasury bonds, corporate bonds.

$$P = C/(1+i) + C/(1+i)^2 + \dots + C/(1+i)^n + F/(1+i)^n$$

■ Example

A 10% coupon bond, face value \$1,000, 10 years to maturity. Pays \$100/year for 10 years + \$1,000 at the end.

If purchased at \$1,000 → YTM = **10%** (equals coupon rate)

If purchased at \$900 → YTM = **11.75%** (higher than coupon rate)

Special Case: Consol / Perpetuity

A consol is a bond with NO maturity date that pays a fixed coupon FOREVER.

Originally issued by British Treasury during the Napoleonic Wars.

$$P = C / i \text{ (Consol Price Formula)}$$

$$i = C / P \text{ (Yield of a Consol)}$$

■ Example

Consol pays \$100/year forever. Interest rate =

10%. Price = $\$100 / 0.10 = \$1,000$

If rate rises to 20%: Price = $\$100 / 0.20 = \500 (price FELL!)

This clearly shows the inverse relationship between price and interest rate.

Type 4: Discount Bond (Zero-Coupon Bond)

Sold BELOW face value (at a discount). NO coupon payments are made. The investor profits purely from the price increase at maturity.

$$i = (F - P) / P \text{ (YTM of Discount Bond)}$$

■ Example

A T-bill with face value \$1,000, purchased today for

\$900. YTM = $(\$1,000 - \$900) / \$900 = \$100 / \$900 =$

11.1%

Examples: U.S. Treasury bills, U.S. savings bonds, long-term zero-coupon bonds.

Quick Comparison of the Four Instruments

Instrument	Pays Interest?	Returns Principal?	Example
Simple Loan	Yes, at end	Yes, at maturity	Commercial loans
Fixed-Payment	Yes, in each payment	Yes, spread over payments	Mortgages, car loans
Coupon Bond	Yes, annually (coupon)	Yes, face value at maturity	Treasury bonds
Discount Bond	No (zero coupon)	Yes, face value at maturity	Treasury bills

4. Yield To Maturity (YTM) – The Gold Standard

Definition

Yield to Maturity (YTM) is the interest rate that equates the present value of all future payments from a debt instrument with its price (value) today.

It is the MOST ACCURATE measure of interest rates. When economists say 'interest rate', they mean YTM.

The Central Principle:

Today's Price = Present Value of ALL Future Payments

4.1 YTM for Simple Loan

For a simple loan, YTM equals the simple interest rate – they are the same thing!

$$i = (\text{Repayment} - \text{Principal}) / \text{Principal}$$

Example

Loan: \$100 today, repay \$110 in 1 year.

$$i = (\$110 - \$100) / \$100 = 10\%$$

Simple interest rate = YTM = 10% ✓

4.2 YTM for Fixed-Payment Loan

$$LV = FP/(1+i) + FP/(1+i)^2 + \dots + FP/(1+i)^n$$

Example

Loan value LV = \$1,000; Payment FP = \$126/year; n = 25

years. Solve for i using financial calculator → i = 12%

4.3 YTM for Coupon Bond

$$P = C/(1+i) + C/(1+i)^2 + \dots + C/(1+i)^n + F/(1+i)^n$$

Three Key Facts About Coupon Bonds:

- **Fact 1:** When bond price = face value → YTM = Coupon Rate
- **Fact 2:** Price and YTM are NEGATIVELY related (price ↑ → YTM ↓, price ↓ → YTM ↑)
- **Fact 3:** When bond price < face value → YTM > Coupon Rate

Yield to Maturity vs. Bond Price (10% Coupon, 10-year bond, FV=\$1,000):

Bond Price (\$)	Yield to Maturity (%)	Relationship to Coupon Rate
1,200	7.13%	YTM < Coupon Rate (bond at premium)
1,100	8.48%	YTM < Coupon Rate (bond at premium)
1,000	10.00%	YTM = Coupon Rate (at par)
900	11.75%	YTM > Coupon Rate (bond at discount)

800	13.81%	YTM > Coupon Rate (bond at discount)
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■ Why Bond Price Falls When Interest Rate Rises

When interest rate (i) rises, the denominators in the PV formula $(1+i)^n$ get LARGER.

Larger denominators → SMALLER present values of future payments → LOWER bond price.

This is the fundamental inverse relationship: **Price** ↑ ■ **YTM** ↓

4.4 YTM for Discount Bond

$$i = (F - P) / P$$

■ Example

Treasury bill: Face Value F = \$1,000; Purchase Price P = \$900.

YTM = $(\$1,000 - \$900) / \$900 = 11.1\%$

If price rises to \$950: YTM = $(\$1,000 - \$950) / \$950 = 5.3\%$ (YTM fell as price rose)

■ Special Case: Negative Interest Rates (Japan, 1998)

In November 1998, Japanese 6-month T-bills had a NEGATIVE interest rate of -0.004%!

Why? Large investors preferred T-bills over cash because T-bills are stored electronically and in larger denominations – even at a small cost.

This shows that while negative rates are extremely unusual, they can occur.

5. Other Measures Of Interest Rates

Because YTM can be difficult to calculate, two simpler (but less accurate) measures are commonly used in practice.

5.1. Current Yield

$$\text{Current Yield (i_c)} = C / P$$

- C = Yearly coupon payment; P = Current price of bond
- This is the same formula as for a consol – so it is EXACT for consols.
- For long-term bonds (20+ years), it is a GOOD approximation of YTM.
- For short-term bonds, it can be very misleading.

■ Example

10% coupon bond, Face Value = \$1,000.

If price = \$1,000: Current Yield = $\$100/\$1,000 = 10\%$ (same as

YTM) If price = \$1,100: Current Yield = $\$100/\$1,100 = 9.09\%$

(YTM = 8.48%)

Both move in the same direction, but current yield overestimates YTM for short-term bonds.

Rules for Current Yield as an Approximation:

Condition	Accuracy of Current Yield
Bond price close to par + long maturity	■ Very Good Approximation
Bond price close to par + short maturity	■ ■ Poor Approximation
Bond price far from par + short maturity	■ Very Misleading
Consol (perpetuity)	■ Exact Measure

5.2. Yield on a Discount Basis (Discount Yield)

$$i_{db} = [(F - P) / F] \times [360 / \text{days to maturity}]$$

Two Quirks of This Formula:

- **Quirk 1:** Uses percentage gain on FACE VALUE $(F-P)/F$, not on purchase price $(F-P)/P$.
- **Quirk 2:** Treats a year as 360 days, not 365 days.
- Both quirks cause the discount yield to **UNDERSTATE** the true YTM.
- The longer the maturity, the greater the understatement.

■ Example

T-bill: Face value \$1,000; Purchase price \$900; 1 year to maturity.

Discount Yield = $[(\$1,000 - \$900)/\$1,000] \times (360/365) = 0.10 \times 0.986 = 9.9\%$

But the YTM (true measure) = $(\$1,000 - \$900)/\$900 = 11.1\%$

Discount yield understates YTM by more than 1 percentage point!

■ Key Takeaway for Both Alternative Measures

Both Current Yield and Discount Yield are **imperfect** measures of the true interest rate (YTM). However, they always **move in the same direction** as YTM.

If Current Yield rises → YTM has also risen. If Discount Yield falls → YTM has also fallen. A change in either always signals a change in the same direction for YTM.

6. Distinction Between Interest Rates & Returns

■ Critical Point

The **INTEREST RATE** (YTM) is not the same as the **RATE OF RETURN!** Interest rate is set when you **BUY** the bond.

Rate of Return depends on what happens to bond **PRICES** while you hold it.

Rate of Return Formula

$$RET = (C + P_{t+1} - P_t) / P_t$$

$$RET = \text{Current Yield (i}_c\text{)} + \text{Rate of Capital Gain (g)}$$

Symbol	Meaning
RET	Rate of return over the holding period
C	Coupon payment
P _t	Price at time t (when you buy)
P _(t+1)	Price at time t+1 (when you sell)
g	Rate of capital gain = $(P_{(t+1)} - P_t) / P_t$

■ Example – Surprising Return

Buy a 10% coupon bond at \$1,000 (YTM = 10%). Receive \$100 coupon payment.

Sell after 1 year for \$1,200.

Capital Gain = \$1,200 – \$1,000 = \$200

Return = $(\$100 + \$200) / \$1,000 = \mathbf{30\%}$ ← much higher than the 10%

YTM! Lesson: Bond prices changed, so return ≠ interest rate.

One-Year Returns When Interest Rates Rise from 10% to 20%

Years to Maturity	Initial YTM (%)	Initial Price (\$)	Price Next Year (\$)	Capital Gain (%)	Return (%)
30	10	1,000	503	–49.7	–39.7
20	10	1,000	516	–48.4	–38.4
10	10	1,000	597	–40.3	–30.3
5	10	1,000	741	–25.9	–15.9
2	10	1,000	917	–8.3	1.7

1	10	1,000	1,000	0.0	10.0
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Five Key Findings from the Table:

- The **only bond where Return = YTM** is when holding period = maturity (1-year bond in example).
- A rise in interest rates causes **capital losses** on long-term bonds.
- **Longer maturity** □ bigger price change for a given change in interest rates.
- **Longer maturity** □ lower return when interest rates rise.
- Even with a 10% initial YTM, a 30-year bond can have a **negative return of –39.7%!**

7. Maturity & Interest-Rate Risk

■ Definition: Interest-Rate Risk

Interest-Rate Risk is the riskiness of an asset's return due to changes in interest rates.

It is especially important for **long-term bonds**, where capital gains and losses can be very large. Short-term bonds (maturity = holding period) have **NO interest-rate risk**.

Why Long-Term Bonds Are Risky:

- Their prices fluctuate dramatically with interest rate changes (20%+ swings are common for 20+ year bonds).
- Capital losses can wipe out all coupon income and more.
- **Long-term bonds are NOT safe assets for short holding periods.**

Why Short-Term Bonds Are Safe:

- When holding period = maturity, the price at the end is already fixed at face value.
- Interest rate changes cannot affect the final price.
- Return always equals the YTM known at purchase.

■■ Reinvestment Risk

If your holding period is **LONGER** than the bond's maturity, you face reinvestment risk.

When the short-term bond matures, you must reinvest proceeds at an uncertain future rate.

If rates fall → you reinvest at a lower rate (bad outcome).

If rates rise → you reinvest at a higher rate (good outcome).

8. Real Vs. Nominal Interest Rates (Fisher Equation)

Two Types of Interest Rates:

Type	Definition	Accounts for Inflation
Nominal Interest Rate (i)	The stated interest rate, not adjusted for inflation.	■ No
Real Interest Rate (i_r)	Adjusted for expected inflation. Reflects the true cost of Borrowing.	Yes

The Fisher Equation:

$$i = i_r + \pi^e \text{ (Fisher Equation)}$$

$$i_r = i - \pi^e \text{ (Real Interest Rate)}$$

Where π^e = expected inflation rate

■ Example 1 – Positive Real Rate

Nominal rate $i = 5\%$, Expected inflation $\pi^e = 3\%$
Real rate $i_r = 5\% - 3\% = \mathbf{2\%}$

You gain 2% more purchasing power per year – a POSITIVE real return.

■ Example 2 – Negative Real Rate

Nominal rate $i = 8\%$, Expected inflation $\pi^e = 10\%$
Real rate $i_r = 8\% - 10\% = \mathbf{-2\%}$

You actually LOSE 2% purchasing power – a NEGATIVE real return. Lender is worse off; borrower benefits (repays in cheaper dollars).

Why Real Interest Rate Matters More:

- The real rate reflects the **true cost of borrowing**.
- It is a better guide to **incentives to borrow and lend**.
- In the 1970s in the U.S., nominal rates were HIGH but real rates were often **NEGATIVE** – meaning borrowing was actually cheap!
- Nominal rates made it LOOK expensive to borrow, but real rates showed it was actually cheap.

■ TIPS – Treasury Inflation Protection Securities (from 1997)

Since January 1997, the U.S. Treasury issues TIPS – indexed bonds whose interest and principal adjust for inflation. These bonds provide a **direct, observable measure** of the real interest rate.

By subtracting TIPS yield from nominal T-bond yield → you get **expected inflation**.

Example (Jan 2003): 10-yr T-bond = 3.84%; 10-yr TIPS = 2.19% → Implied inflation =

1.65%

9. Key Terms Quick Reference

Term	Definition
Basis Point	1/100th of a percentage point. E.g. rate rises from 5.00% to 5.01% = 1 basis point.
Consol / Perpetuity	A bond with no maturity that pays fixed coupon forever. Price = C/i .
Coupon Bond	Pays fixed coupon payments annually + face value at maturity. E.g. Treasury bonds.
Coupon Rate	Annual coupon payment ÷ face value. E.g. $\$100/\$1,000 = 10\%$.
Current Yield	Coupon payment ÷ bond price. Approximation of YTM for long-term bonds.
Discount Bond	Zero-coupon bond bought below face value. Profit comes from price appreciation.
Face Value (Par Value)	The amount repaid at maturity. Usually \$1,000 for bonds.
Fixed-Payment Loan	Loan repaid in equal periodic installments (principal + interest). E.g. mortgage.
Indexed Bond	Bond whose payments are adjusted for inflation. E.g. TIPS.
Interest-Rate Risk	Risk that interest rate changes will cause capital gains/losses on a bond.
Nominal Interest Rate	Stated interest rate; does NOT adjust for inflation.
Present Value (PV)	Today's value of a future payment. $PV = FV / (1+i)^n$.
Rate of Capital Gain	Change in bond price divided by initial price: $(P_{t+1} - P_t) / P_t$.
Real Interest Rate	Nominal rate minus expected inflation. Reflects true cost of borrowing.
Rate of Return	Total return: current yield + rate of capital gain.
Simple Loan	Principal lent, repaid at maturity with interest. Simple loan rate = YTM.
Yield on Discount Basis	$[(F - P) / F] \times (360/\text{days})$. Used for T-bills; understates YTM.
Yield to Maturity (YTM)	The rate that equates PV of future payments to today's price. Most accurate measure.

10. Chapter Summary

1. Present Value	A dollar in the future is worth less today. $PV = FV/(1+i)^n$. This concept allows comparing instruments with different payment timings.
2. Four Debt Instruments	Simple Loan, Fixed-Payment Loan, Coupon Bond, Discount Bond – each has different payment structures but can all be analyzed using PV.
3. Yield to Maturity (YTM)	The most accurate interest rate measure. It equates PV of future payments to today's price. Bond prices and YTM are always NEGATIVELY related.
4. Other Rate Measures	Current Yield = C/P (good for long-term bonds). Discount Yield uses face value and 360-day year (always understates YTM). Both move in same direction as YTM.
5. Returns vs. Interest Rates	Return = current yield + capital gain/loss. Return = YTM only when holding period = maturity. Long-term bonds carry substantial interest-rate risk.
6. Real vs. Nominal Rates	Real Rate = Nominal Rate – Expected Inflation (Fisher Equation). Real rate better reflects true borrowing cost. High nominal rates $\neq$ high real rates (e.g. U.S. 1970s).

Formula Sheet

Concept	Formula
Present Value	$PV = FV / (1 + i)^n$
YTM – Simple Loan	$i = (\text{Repayment} - \text{Principal}) / \text{Principal}$
YTM – Fixed Loan	$LV = \sum [FP / (1+i)^t] \text{ for } t=1 \text{ to } n$
YTM – Coupon Bond	$P = \sum [C/(1+i)^t] + F/(1+i)^n$
Consol Price	$P = C / i$
Consol Yield	$i = C / P$
YTM – Discount Bond	$i = (F - P) / P$
Current Yield	$i_c = C / P$
Discount Yield	$i_{db} = [(F-P)/F] \times [360/\text{days}]$
Rate of Return	$RET = (C + P_{(t+1)} - P_t) / P_t$
Return Decomposed	$RET = \text{Current Yield} + \text{Capital Gain Rate}$
Fisher Equation	$i = i_r + \pi^e$
Real Interest Rate	$i_r = i - \pi^e$